

Assessing the Accuracy of the Black-Scholes Model on Nifty50 Index Options

A back-test against real NSE options data (Q1 2024)
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Abstract

We test how accurately the Black-Scholes-Merton (BSM) model prices Nifty50 index options using **43,817 real traded quotes** drawn from the official NSE F&O bhavcopy over 60 trading days (2024-01-01 to 2024-03-28). Options are priced in the Black-76 forward form, with the forward and discount factor recovered directly from put-call parity. We evaluate three ex-ante volatility inputs—trailing realized volatility, EWMA, and GARCH(1,1)—and benchmark flat-volatility BSM against a smile-fitted model. BSM tracks the level of option prices closely ($R^2 > 0.99$) but exhibits an economically meaningful pricing error (RMSE $\approx$ Rs. 51, MAPE $\approx$ 37%), systematically **under-pricing** options. Errors concentrate out-of-the-money, where a pronounced volatility smile (median skew -0.21, convexity +3.5) violates BSM's constant-volatility assumption. Feeding a GARCH forecast cuts RMSE by 22%, and fitting the smile cuts it by 41%—localizing BSM's inaccuracy to its flat-volatility assumption rather than the pricing formula itself.

1. Introduction and Objective

The Black-Scholes-Merton model is the foundational framework for option pricing, yet it rests on assumptions—constant volatility, lognormal returns, frictionless continuous hedging—that are known to be violated in real markets. This project quantifies *how large* the resulting pricing errors are for the Nifty50 index options market, *where* they concentrate, and *why* they arise. Rather than treating BSM as simply 'right' or 'wrong', we decompose its error into (i) the choice of volatility input and (ii) the structural constant-volatility assumption, and we measure the contribution of each.

2. Data

The dataset is built from the **official NSE Futures & Options daily bhavcopy** (settlement files), filtered to Nifty index options (INSTRUMENT = OPTIDX, SYMBOL = NIFTY). We retain only contracts that actually traded on the day (CONTRACTS > 0), so every price is a genuine transaction price rather than a stale settlement mark. This yields **43,817 option quotes** across 60 trading days and 20 expiries (2024-01-01 to 2024-03-28), spanning strikes 14000–27000 and maturities of 1–120 days.

Because Nifty lists weekly options but only monthly futures, we do not rely on a matched futures contract for the underlying. Instead, for every (date, expiry) we recover the **forward price** from put-call parity, $C(K) - P(K) = D \cdot (F - K)$, using liquidity-weighted near-the-money strike pairs at a fixed risk-free rate of 6.9% (the Q1-2024 91-day T-bill yield). This CBOE/VIX-style construction needs no dividend assumption and is internally consistent with the prices under test. The recovered forwards reproduce the traded Nifty level to within a few points, and at-the-money call and put prices coincide, confirming parity holds.

3. Methodology

Pricing. Index options are priced with Black-76, the forward form of Black-Scholes: a call is $D \cdot [F \cdot N(d1) - K \cdot N(d2)]$ with $d1 = [\ln(F/K) + \frac{1}{2}\sigma^2 T] / (\sigma\sqrt{T})$. This is algebraically BSM with the dividend yield set equal to the rate, and it isolates volatility as the only free input.

Volatility inputs (all ex-ante). (1) *Realized*: 20-day trailing close-to-close volatility, annualized. (2) *EWMA*: RiskMetrics exponentially-weighted volatility ($\lambda = 0.94$), which reacts faster to shocks. (3) *GARCH(1,1)*: a one-day-ahead conditional-volatility forecast estimated by maximum likelihood on the underlying return series.

Implied volatility & smile. We invert Black-76 numerically (vectorized Newton-Raphson with a bisection fallback) to obtain each option's implied volatility, successfully recovering IV for 93% of quotes. For each expiry we fit IV as a quadratic in log-moneyness, $IV(x) = a + b \cdot x + c \cdot x^2$, whose slope b measures skew and curvature c measures the smile.

Accuracy metrics. We report RMSE, MAE, MAPE, mean signed error and R^2 , overall and bucketed by moneyness and maturity, with bootstrapped 95% confidence intervals on RMSE (2,000 resamples).

4. Results

4.1 Accuracy by volatility input

Vol. input	RMSE (Rs.)	95% CI	MAE	MAPE %	Mean err.	R^2
realized	50.8	[50, 52]	30.8	37.5	-9.3	0.996
ewma	40.5	[40, 41]	23.6	33.1	-8.8	0.997
garch	39.5	[39, 40]	23.4	33.8	-10.0	0.998

All three inputs price the *level* of option value almost perfectly ($R^2 > 0.99$), because deep in-the-money options are dominated by intrinsic value. The economically relevant differences appear in RMSE and MAPE: the **GARCH forecast is the most accurate**, lowering RMSE by 22% versus naive realized volatility. The negative mean signed error under every model shows BSM **systematically under-prices** Nifty options—consistent with a variance risk premium, since at-the-money implied volatility (14%) sits above realized volatility.



Figure 1. Pricing error (RMSE) by volatility input and for the smile-fitted benchmark.

4.2 The volatility smile

Backing out implied volatility per option reveals a pronounced, asymmetric smile: out-of-the-money puts trade at far higher implied volatilities than at-the-money options, while a single flat BSM volatility (dashed line) sits below almost the entire curve. This is the structural reason BSM under-prices—it cannot simultaneously match all strikes with one volatility.

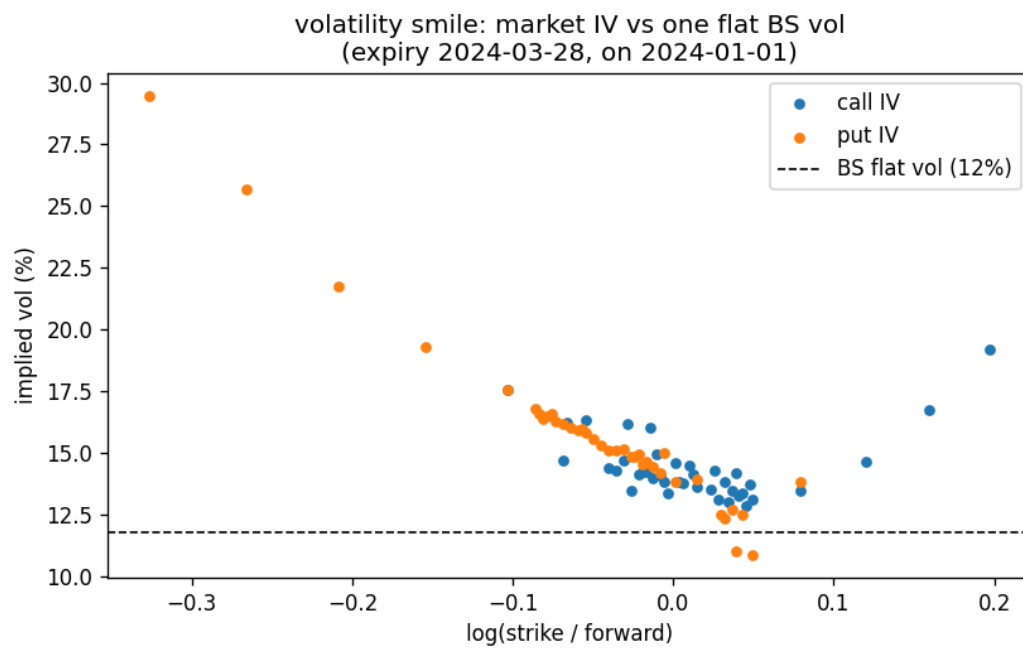


Figure 2. Market implied volatility vs BSM's constant-volatility assumption (nearest liquid expiry).

4.3 Where BSM fails: error by moneyness and maturity

Moneyneess	n	RMSE	MAE	MAPE %	R ²
ATM	6,190	60.4	44.3	15.3	0.828
ITM	13,569	63.6	40.4	4.3	0.996
OTM	24,058	38.4	21.9	61.9	0.823

In-the-money options, dominated by intrinsic value, are priced accurately (single-digit MAPE). The error explodes for out-of-the-money options (MAPE well above 50%), exactly where the smile is steepest and small volatility misspecifications translate into large percentage price errors.

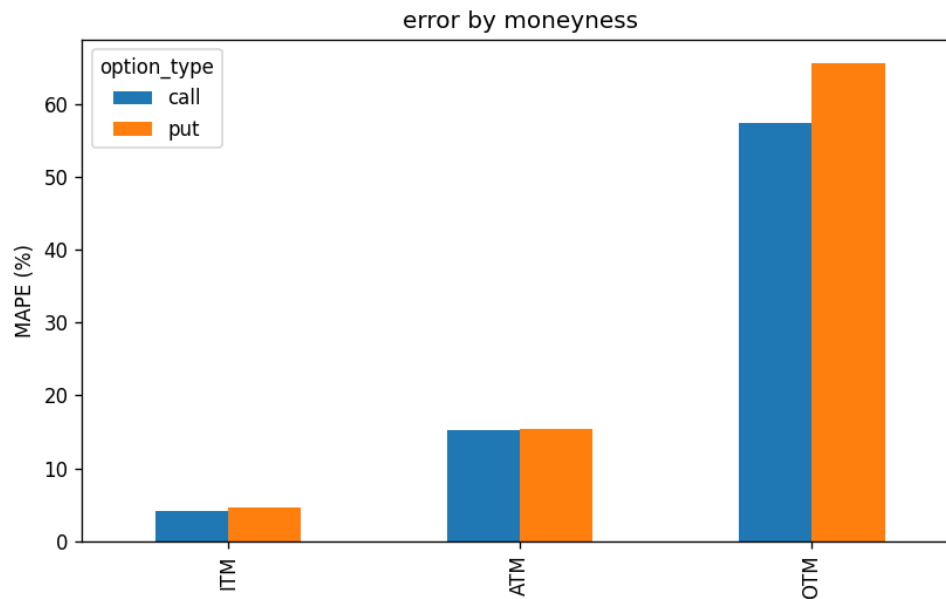


Figure 3. BSM percentage error by moneyness, calls vs puts.

4.4 Isolating the cause: the smile-fitted benchmark

To separate the formula from the flat-volatility assumption, we reprice every option using its expiry's fitted smile rather than a single volatility. RMSE falls from Rs. 50.8 to Rs. 30.2—a **41% reduction**—and R² rises to 0.999. Because the only change is allowing volatility to vary with strike, this clearly attributes the bulk of BSM's error to its constant-volatility assumption, not to the option-pricing formula.

5. Discussion

Three findings stand out. First, BSM is structurally sound as a pricing map—given the right volatility it explains virtually all cross-sectional price variation. Second, its accuracy is governed almost entirely by the volatility input: a forward-looking GARCH forecast materially outperforms backward-looking realized volatility. Third, the residual error is not random but structural—the volatility smile—and can be almost entirely removed by letting volatility depend on strike. Together these results reframe 'is BSM accurate?' into the more useful question of 'how should its volatility input be specified?'

6. Limitations

The study covers one quarter (Q1 2024) of a single index; results should be confirmed over multiple volatility regimes. Prices are daily closes, so call and put legs are not perfectly synchronous; the parity-based forward mitigates but does not eliminate this. The risk-free rate is held constant, and American-exercise and transaction-cost effects are ignored (reasonable for European-style, liquid index options). GARCH is estimated in-sample; a strict walk-forward re-estimation would be the natural next step.

7. Conclusion

Black-Scholes is an accurate *pricing engine* but an incomplete *volatility model*. On 43,817 real Nifty50 quotes it explains the level of option prices ($R^2 > 0.99$) yet systematically under-prices by design, with error concentrated in out-of-the-money strikes. Improving the volatility input (GARCH) and, above all, replacing the flat-volatility assumption with a fitted smile recover most of the lost accuracy—cutting RMSE by 22% and 41% respectively. The practical lesson is that the model's error lives in its volatility assumption, and that is where effort should be spent.

References

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